

To develop and fine-tune a **BERT-based Sentiment Analysis model** that classifies text reviews as **positive** or **negative** using the IMDb dataset.

Model Overview

- **Model Used:** bert-base-uncased (Bidirectional Encoder Representations from Transformers)
- **Type:** Pre-trained transformer model fine-tuned for binary sentiment classification
- **Framework:** Hugging Face Transformers with PyTorch backend
- **Task:** Sequence classification using BertForSequenceClassification

Dataset

- **Source:** IMDb Movie Reviews dataset (Hugging Face datasets library)
- **Size Used:** 3,000 samples for training and 500 samples for testing (subset for quick training)
- **Classes:**
 - LABEL_1 → Positive sentiment
 - LABEL_0 → Negative sentiment

Implementation Steps

1. **Data Loading:** IMDb dataset loaded and subset created for faster training.
2. **Preprocessing:**
 - Text tokenized using BertTokenizer with max sequence length = 128.
 - Attention masks applied for padded sequences.
3. **Model Fine-tuning:**
 - Base model: bert-base-uncased
 - Optimized using AdamW optimizer (default in Trainer).
 - Batch size = 4, Epochs = 1, Learning rate = 2e-5

4. Evaluation:

- Predictions obtained manually using `Trainer.predict()`
- Metrics calculated using scikit-learn functions

Results

Metric Value

Accuracy 85.4 %

Precision 81.4 %

Recall 91.0 %

F1-Score 85.9 %

The model performed well even with a small training subset and a single epoch.

Custom Sentence Predictions

Sentence	Predicted Sentiment	Confidence
<i>The movie was absolutely amazing and emotional!</i>	Positive (LABEL_1)	99.28 %
<i>I hated the plot, it was boring and predictable.</i>	Negative (LABEL_0)	98.85 %
<i>The acting was great but the story was weak.</i>	Negative (LABEL_0)	95.36 %
<i>Terrible. I walked out halfway through.</i>	Negative (LABEL_0)	98.15 %

Model Artifacts

- Fine-tuned Model saved at: `./sentiment-bert-model`
- Tokenizer saved at: `./sentiment-bert-tokenizer`
- Logs stored in: `./logs/`
- Reproducible Colab notebook used for training and evaluation

Conclusion

The fine-tuned BERT model successfully identifies sentiment polarity in movie reviews with high accuracy and balanced precision–recall.

The implementation demonstrates the power of **transformer-based NLP models** for sentiment analysis tasks.

For extended performance:

- Increasing training epochs or dataset size can raise accuracy to ~90%.
- A lighter model such as **DistilBERT** can be used for faster execution with similar results.